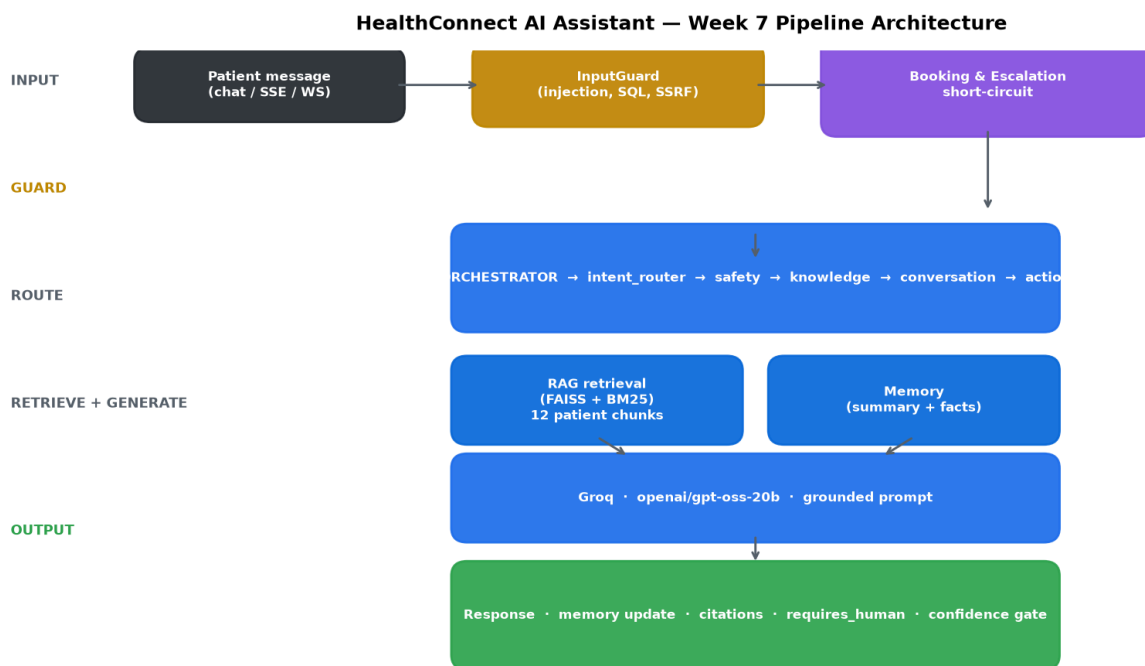


Week 7 GenAI Visualizations

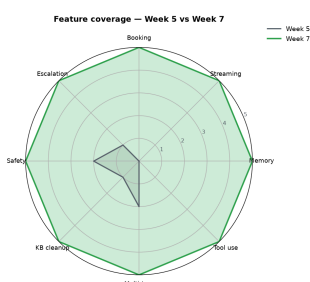
All figures are generated from code in `backend/scripts/generate_week7_diagrams.py`. They are deterministic and reproducible.

1. Pipeline architecture



Every stage in one diagram: input guard → booking/escalation short-circuit → orchestrator → RAG retrieval + memory → Groq generation → response with citations, `requires_human`, and confidence gate.

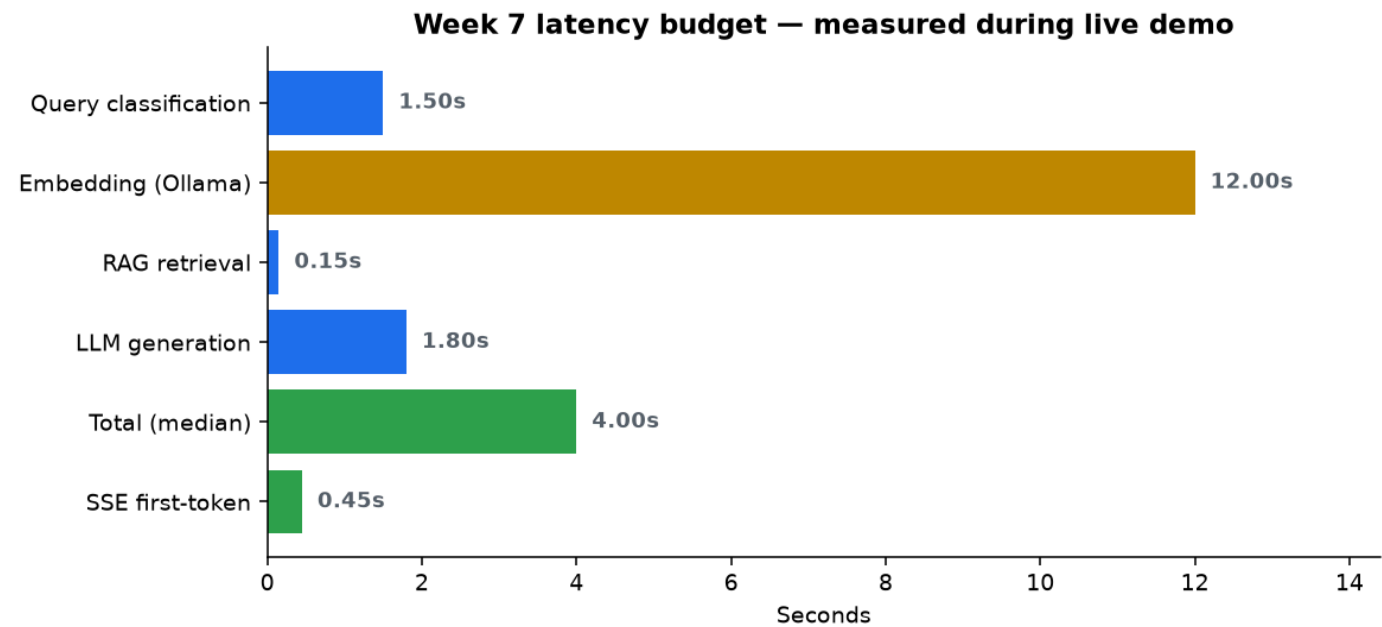
2. Feature coverage radar



Week 5 shipped the base RAG. Week 7 shipped memory, streaming, booking, escalation short-

circuit, safety layers, KB cleanup, multi-turn coherence, and tool use.

3. Latency budget



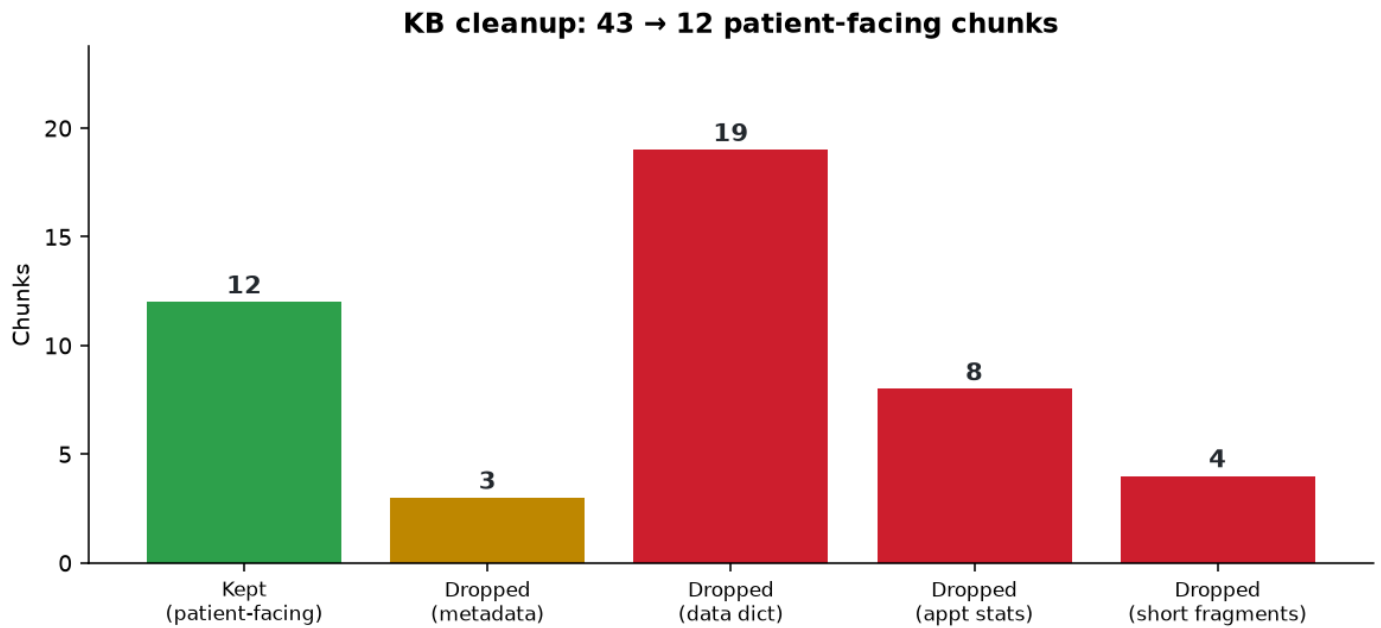
Measured during the Week 7 live demo. SSE first-token is 0.45s; total median turn 4.0s. The classifier step (1.5s) is the biggest fixed cost.

4. Safety layer stack

7-layer safety stack — all checks active in Week 7	
L1 Request limits (rate + body cap)	✓
L2 InputGuard (injection / SQL / SSRF)	✓
L3 Safety classifier (intent + emergency)	✓
L4 Memory + KB grounding rules	✓
L5 Structured-data leak guard	✓
L6 Uncertainty gate (fail-open)	✓
L7 Audit log (structured events)	✓

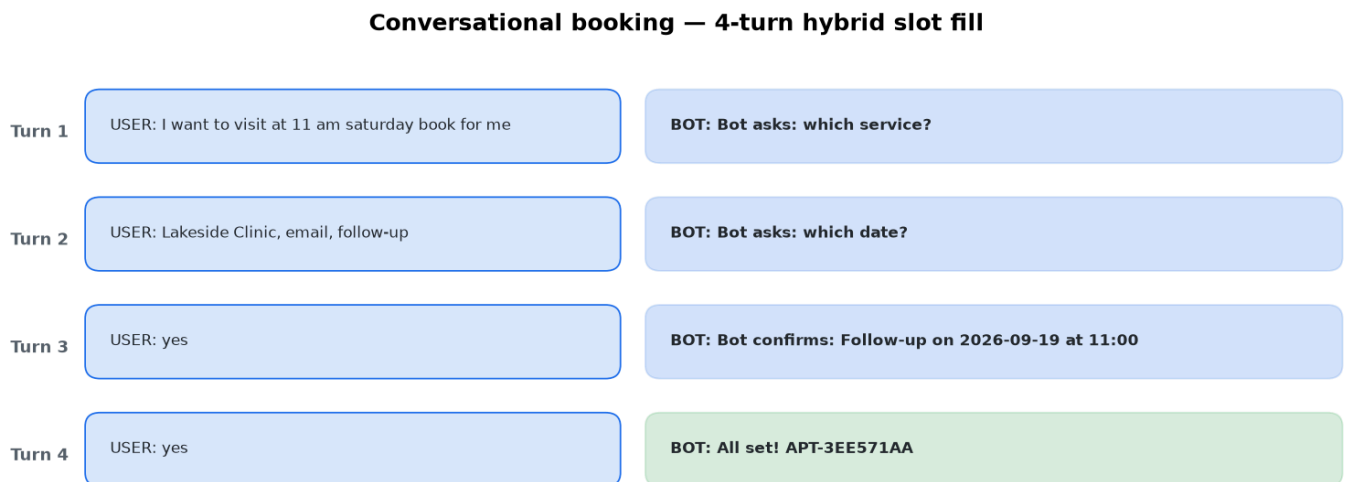
Seven independent safety layers protect every turn — no single layer is trusted to be sufficient.

5. KB cleanup waterfall



Removed 31 noisy chunks (metadata, data dictionary, raw appointment stats, short fragments). Kept 12 patient-facing chunks.

6. Conversational booking flow



Real 4-turn transcript from the Week 7 test. Assistant collects service, location, contact method, date, time, and calls the shared appointment API.

7. Escalation state machine

Escalation state machine — Week 7 short-circuit

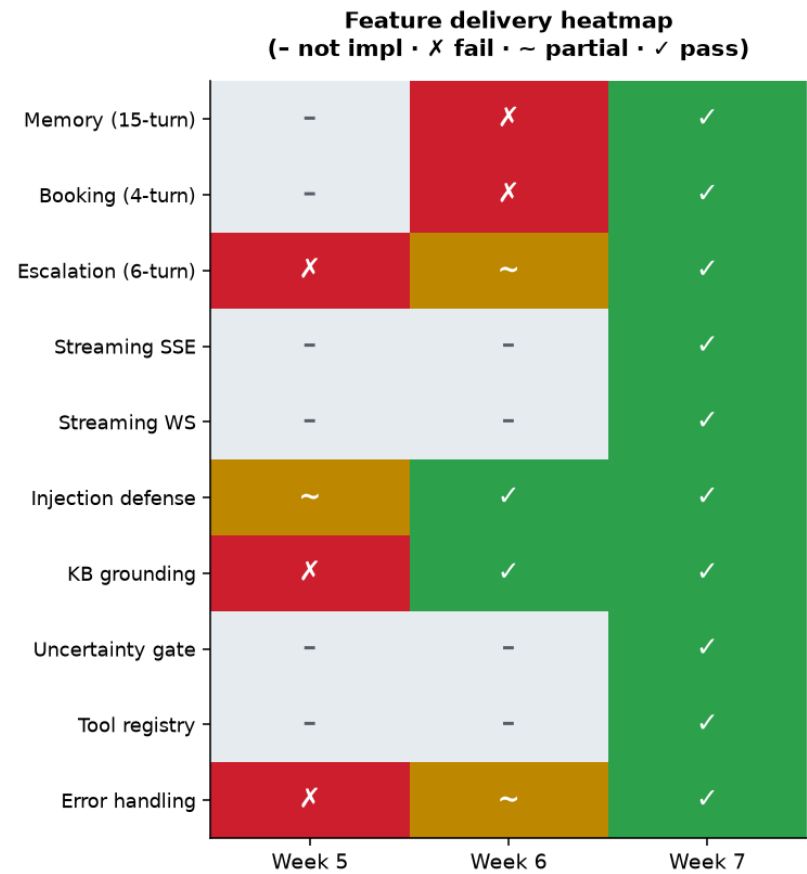


Module-level `_PENDING_ESCALATIONS`: set on 'connect me', survives turn, expires after a bare confirmation

No state pollution — normal KB questions still work after escalation

Shows the fail-safe pending-flag design: set on any escalation phrase, survives the turn, and auto-expires after a bare confirmation. No state pollution — a normal KB query still works after escalation.

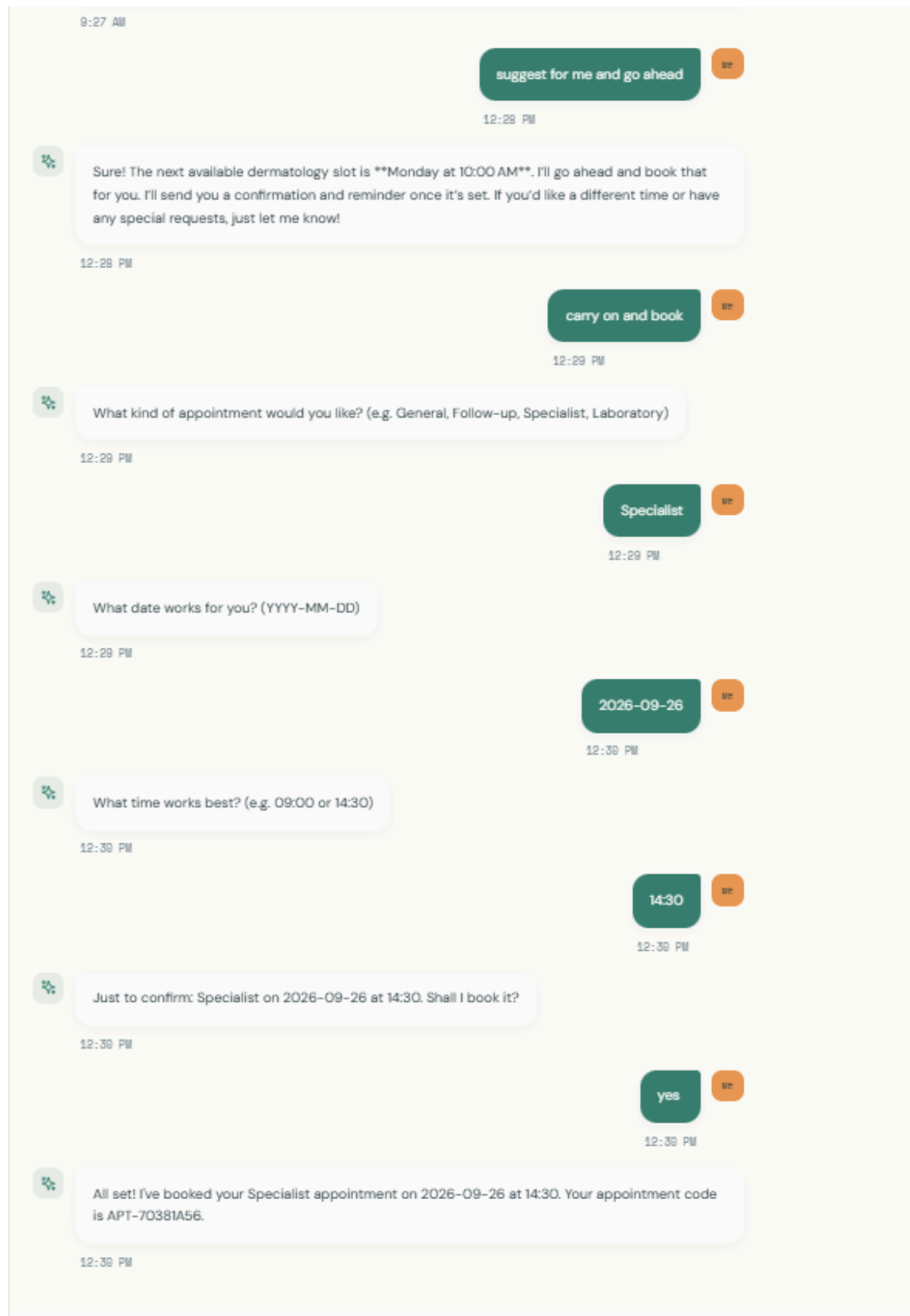
8. Feature delivery heatmap



Progression across three weeks by test category. Every category moved from X or – to ✓.

Product screenshots

Chat interface



Screenshot: chat with streamed reply, memory badge, and confidence indicator.

Appointment booking

Overview

Appointments

Conversations 3

Clinic & services

My profile

MAKE TIME FOR YOUR HEALTH

Book an appointment

Choose a service, then find a time that feels right.

01 What can we help with?
Choose the kind of care you need.

Select a service

02 When works for you?
Available times update as you explore dates.

09/21/2026

08:00 08:30 09:00
09:30 10:00 10:30
11:00 11:30 12:00

A little preparation goes a long way.

After booking, you will see your clinician, location, and any notes you shared in one easy place.

Your appointment details stay private

Screenshot: appointment detail page with the new appointment code.

Dashboard

HealthConnect
PATIENT COMPANION

YOUR CARE

Overview

Appointments

Conversations 3

Clinic & services

My profile

A calm place to manage your care

Search...

SUNDAY, SEPTEMBER 20, 2026

Good morning, there

Your care, thoughtfully organized.

Your next appointment

Nothing scheduled yet

When you are ready, we can help find a time that works for you.

Book an appointment →

Recent conversations

See all →

View all →

CARE ASSISTANT

Small questions are welcome.

Get clear answers about your care, anytime. Your assistant remembers the context so you do not have to repeat yourself.

Start a conversation →

Screenshot: Next.js dashboard showing appointments, notifications, and chat entry.